Year 10 problem solving

Name _____ Date _____

Answer all questions. 35 marks. Diagrams are NOT accurately drawn, unless otherwise indicated.

WORKED EXAMPLE

Liam sells 200 tickets for a charity quiz night.

Liam receives a total of £1320 from selling the tickets.

$\frac{3}{10}$ of the tickets sold are child tickets.

The rest of the tickets sold are adult tickets.

The cost of an adult ticket is £7.50

(a) Work out the number of child tickets sold.

(1 mark)

(b) Work out the cost of a child ticket.

(3 marks)

(a)

B1 $\frac{3}{10} \times 200 = 60$

Find the fraction of the total number of tickets.

Answer: 60

(b)

P1 *200 − 60 = 140 adult tickets, 140 × £7.50 = £1050*

Work out the money from adult tickets.

P1 *£1320 − £1050 = £270*

Whatever is left must have come from the child tickets.

A1 *£270 ÷ 60 = £4.50*

Share the child money between the child tickets.

Answer: £4.50

FOUNDATIONAL

1 Ibrahim sells 800 tickets for a school concert.

Ibrahim receives a total of £4440 from selling the tickets.

$\frac{3}{10}$ of the tickets sold are child tickets.

The rest of the tickets sold are adult tickets.

The cost of an adult ticket is £6

(a) Work out the number of child tickets sold.

(1 mark)

(b) Work out the cost of a child ticket.

(3 marks)

Non-calculator

2 The two-way table shows some information about the desserts chosen at lunch yesterday by 171 students from Year 7 and Year 8.

Each student chose one dessert from apple pie or fruit or ice cream.

	apple pie	**fruit**	**ice cream**	**Total**
Year 7				84
Year 8		45	4	87
Total	82		23	171

(a) Complete the two-way table.

(3 marks)

(b) One of the 171 students is picked at random.

Write down the probability that this person is one of the Year 8 students.

(1 mark)

Non-calculator

3 **(a)** Emeka and Jasmine share some stickers in the ratio 6 : 1

They share 49 stickers altogether.

Work out how many stickers Emeka receives.

(2 marks)

(b) Another time, Emeka and Jasmine share some stickers in the ratio 5 : 2

Emeka receives 18 stickers more than Jasmine.

Work out how many stickers Jasmine receives.

(2 marks)

Non-calculator

CORE

4 The diagram shows the plan of the floor of a stage.

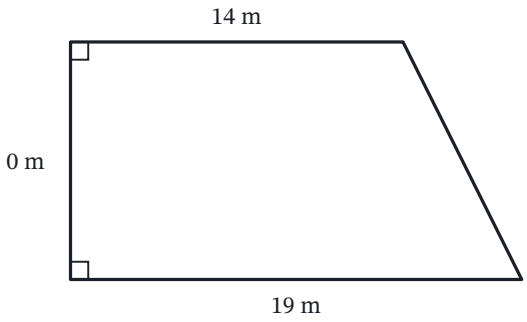


Diagram NOT accurately drawn

The stage manager is going to paint the floor with one coat of paint.

One tin of paint covers an area of 7 m^2

One tin of paint costs £12.50

The stage manager buys the least possible number of tins.

Work out the total cost of the tins that the stage manager buys.

(5 marks)

Non-calculator

5 In a bag, there are only red counters, blue counters and white counters.

There are 66 counters in the bag so that

the number of red counters : the number of blue counters = 1 : 2

and

the number of blue counters : the number of white counters = 3 : 1

Work out how many red counters there are in the bag.

(3 marks)

Non-calculator

6 Kai, Kemi, Ciara and Sofia are four friends.

Here is some information about the heights of these friends.

Kai's height is 180 cm.

Kemi's height is 185 cm.

Ciara's height is greater than Sofia's height.

The median height of these four friends is 174 cm.

The range of the heights of these four friends is 24 cm.

Work out Ciara's height and Sofia's height.

(3 marks)

Non-calculator

STRETCH

7 Mai and Amelia each pay for a bike.

Mai's bike	Amelia's bike
Normal price £1480 Special offer: k% off the normal price	Deposit of £570 plus 12 monthly payments of £45

The amount that Mai pays is the same as the amount that Amelia pays.

Work out the value of k

(4 marks)

Non-calculator

8 The diagram shows an isosceles triangle with base length 19 cm.

The area of the triangle is 177 cm^2

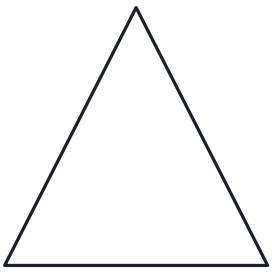


Diagram NOT accurately drawn

Work out the perimeter of the triangle.

Give your answer correct to 3 significant figures.

(4 marks)

Calculator allowed

9 Here is a biased four-sided dice.

When the dice is thrown once, the probabilities that it lands on 1 or 3 are given in the table.

Number	**1**	**2**	**3**	**4**
Probability	0.2		0.2	

The probability that the dice lands on 2 is four times the probability that the dice lands on 4

Kai is going to throw the dice 150 times.

Work out an estimate for the number of times the dice lands on 2

(4 marks)

Non-calculator

Year 10 problem solving: answers

Question 1

(a)

B1 $\frac{3}{10} \times 800 = 240$

Find the fraction of the total number of tickets.

Answer: 240

(b)

P1 $800 - 240 = 560$ adult tickets, $560 \times £6 = £3360$

Work out the money from adult tickets.

P1 $£4440 - £3360 = £1080$

Whatever is left must have come from the child tickets.

A1 $£1080 \div 240 = £4.50$

Share the child money between the child tickets.

Answer: £4.50

Question 2

(a)

B1 Year 8, apple pie: $87 - 45 - 4 = 38$; Total, fruit: $171 - 82 - 23 = 66$

Look for a row or column with only one gap: the numbers must add to the total.

B1 Year 7, apple pie: $82 - 38 = 44$; Year 7, fruit: $66 - 45 = 21$

Each value you find opens up another row or column.

B1 Year 7, ice cream: $23 - 4 = 19$

Check that every row and column adds up.

Answer: Completed table (apple pie, fruit, ice cream, Total) — Year 7: 44, 21, 19, 84; Year 8: 38, 45, 4, 87; Total: 82, 66, 23, 171

(b)

B1 $\frac{87}{171} = \frac{29}{57}$

Probability = number of Year 8 students ÷ total.

Answer: $\frac{29}{57}$

Question 3

(a)

M1 $6 + 1 = 7$ parts, $49 \div 7 = 7$

Add the ratio to find the number of parts, then find one part.

A1 $6 \times 7 = 42$ stickers

Emeka gets 6 parts.

Answer: 42 stickers

(b)

M1 $18 \div 3 = 6$

Emeka has 3 more parts than Jasmine, worth 18.

A1 $2 \times 6 = 12$ stickers

Jasmine gets 2 parts.

Answer: 12 stickers

Question 4

P1 Rectangle: $14 \times 10 = 140$ m²

Split the floor into a rectangle and a triangle.

P1 Triangle base: $19 - 14 = 5$, area: $\frac{1}{2} \times 5 \times 10 = 25$ m²; total 165 m²

Find the missing base of the triangle first.

P1 $165 \div 7 = 23.57...$, so 24 tins

Divide by the area one tin covers and round UP.

P1 $24 \times$ £12.50

Multiply by the price of one.

A1 £300

Include the £ sign.

Answer: £300

Question 5

P1 $1 : 2 = 3 : 6$ and $3 : 1 = 6 : 2$, so $3 : 6 : 2$

Make the blue counters numbers the same (6) so the ratios can be joined.

P1 $66 \div 11 = 6$

Add the parts of the three-part ratio, then find one part.

A1 $3 \times 6 = 18$

The red counters have 3 parts.

Answer: 18

Question 6

P1 Range 24: smallest = 185 − 24 = 161, so Sofia = 161

Use the range to find the unknown at one end of the list.

P1 Median 174: the middle two add up to 348, so Ciara = 348 − 180 = 168

With four values, the median is halfway between the 2nd and 3rd in order.

A1 Ciara 168 cm, Sofia 161 cm

Check in order: 161, 168, 180, 185. Ciara must be greater than Sofia.

Answer: Ciara: 168 cm, Sofia: 161 cm

Question 7

P1 £570 + 12 × £45 = £1110

Amelia pays the deposit plus all of the monthly payments.

P1 £1480 − £1110 = £370 off

So Mai must also pay £1110. Find how much was taken off.

P1 £370 ÷ £1480 × 100

Write the reduction as a percentage of the normal price.

A1 $k = 25$

Check: 25% off £1480 gives £1110

Answer: $k = 25$

Question 8

P1 ½ × 19 × h = 177, so h = 177 × 2 ÷ 19 = 18.63...

Work backwards from the area formula to find the height.

P1 Half the base = 9.5

The height meets the base at its midpoint.

P1 $s = \sqrt{9.5^2 + h^2}$ = 20.9137...

Pythagoras: add the squares. Keep the full value on the calculator.

A1 2 × 20.9137... + 19 = 60.8 cm

The perimeter uses both equal sides. Round at the end.

Answer: 60.8 cm

Question 9

P1 $0.2 + 0.2 = 0.4$, so $1 - 0.4 = 0.6$ for 2 and 4 together

All the probabilities add up to 1.

P1 $4 = x$, $2 = 4x$, so $5x = 0.6$

"Four times" means 4 lots of the other probability.

P1 $x = 0.12$, so $P(2) = 4 \times 0.12 = 0.48$

Solve, then find the probability you need.

A1 $0.48 \times 150 = 72$

Expected number = probability × number of trials.

Answer: 72